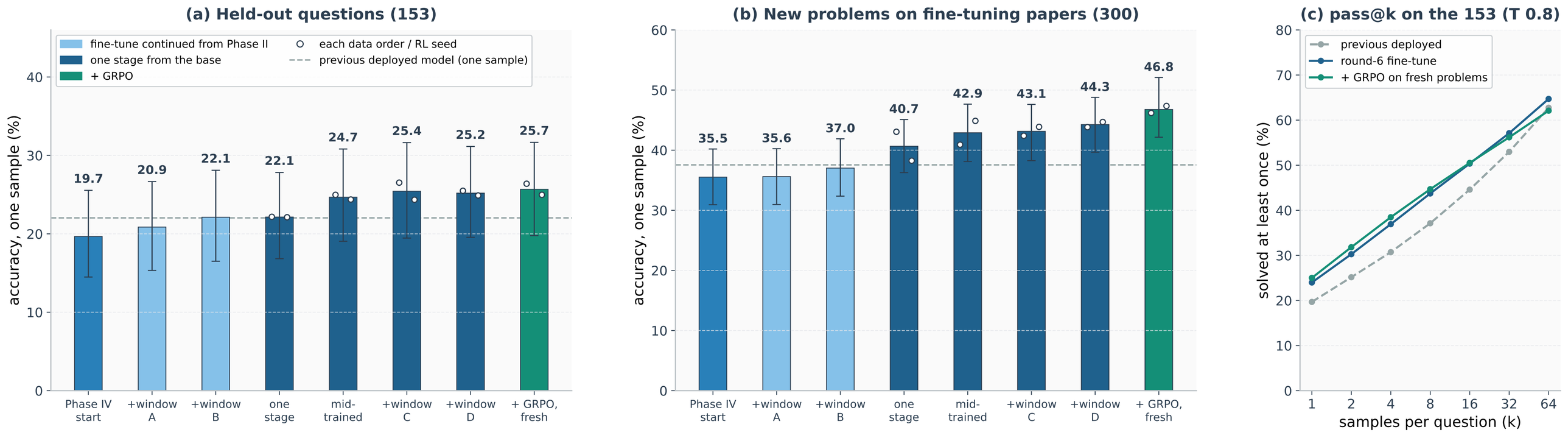
(a) Held-out questions (153)
fine-tune continued from Phase II
one stage from the base
+ GRPO
each data order / RL seed
previous deployed model (one sample)
accuracy, one sample (%)
0
10
20
30
40
19.7
20.9
22.1
22.1
24.7
25.4
25.2
25.7
Phase IV start
+window A
+window B
one stage
mid-trained
+window C
+window D
+ GRPO, fresh
(b) New problems on fine-tuning papers (300)
accuracy, one sample (%)
0
10
20
30
40
50
60
35.5
35.6
37.0
40.7
42.9
43.1
44.3
46.8
Phase IV start
+window A
+window B
one stage
mid-trained
+window C
+window D
+ GRPO, fresh
(c) pass@k on the 153 (T 0.8)
previous deployed
round-6 fine-tune
+ GRPO on fresh problems
solved at least once (%)
0
10
20
30
40
50
60
70
80
1
2
4
8
16
32
64
samples per question (k)